

Problem 1. (20 points)

- a) (**4 pts**) Go to your favorite source of reliable information and look up a value for the mass of the Moon. (Express the value in units of kilograms.)

$$M_{\text{Moon}} = 7.35 \times 10^{22} \text{ kg}.$$

- b) (**10 pts**) Calculate the Schwarzschild radius, r_s , for the Moon, using the equation given above. (Express the result in units of meters.)

$$R_{\text{Schwarzschild, Moon}} = \frac{2GM_{\text{Moon}}}{c^2} = \frac{2 \times 6.67 \times 10^{-11} \text{ m}^3/(\text{kg s}^2) \times 7.35 \times 10^{22} \text{ kg}}{(3 \times 10^8 \text{ m/s}^2)} = 0.00011 \text{ m} = 0.011 \text{ cm}$$

- c) (**6 pts**) In the 2009 movie *Despicable Me*, the arch-criminal Gru uses a “shrink ray” to reduce the Moon to roughly the size of a grapefruit or a softball. Is this shrunken size sufficiently small (assuming that the mass of the Moon has not been changed by shrinking it) for the Moon to become a black hole? Explain your answer.

No, a grapefruit is much larger than a centimeter across.